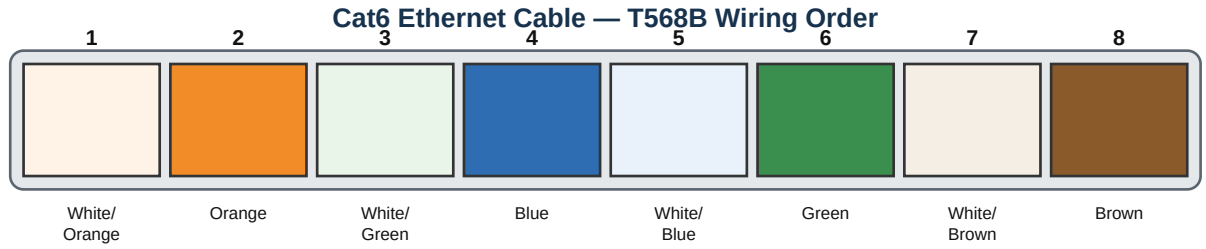
Network Cable Assignment

Name: _____ Class: _____

1. Cat6 Ethernet Cable: T568B and RJ-45 Pins

The T568B standard sets the wire order at the RJ-45 plug. Keep the same standard at both ends for a normal straight-through Ethernet cable.



Pin 1 is on the left when looking at the plug's gold contacts with the clip facing away/down.

T568B order (pins 1 to 8): White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

2. Three Types of Network Cables and Their Uses

Cable type	Example use	Why it is used
Cat5e Ethernet	Wi-Fi router to a POS terminal or laptop	Supports Gigabit Ethernet up to 100 m, suitable for everyday off
Cat6 Ethernet	Gaming console or screen-control PC to a network switch	Supports 1 Gb/s up to 100 m and can support 10 Gb/s over sho
Fiber optic	High-speed link between a venue switch and a server/core switch	Uses light, supports long distances and high bandwidth, and is r

3. Steps to Crimp an RJ-45 Connector

1. Measure and cut the Cat6 cable to the required length.

2. Carefully remove about 2–3 cm of the outer jacket without cutting the inner wires.

3. Untwist the wire pairs only as much as needed, then arrange the eight wires in T568B order.

4. Straighten the wires and trim their ends evenly so they fit inside the RJ-45 plug.

5. Insert all eight wires fully into the plug. Check that the order is correct and the outer jacket reaches under the plug's strain-relief tab.

6. Use an RJ-45 crimping tool to crimp the connector firmly.

7. Repeat at the other end using T568B for a normal straight-through cable.

8. Test the cable with a cable tester or connect devices and check the link.

Two common mistakes to avoid: (1) Mixing up the wire color order. (2) Not pushing all wires fully into the plug or failing to trap the outer jacket under the strain relief.

Video reference: Search YouTube for “How to crimp RJ45 Cat6 cable T568B step by step”. Watch a demonstration and follow the safety instructions for the cutting/crimping tools.

4. Cable Choices for an IPL Screening Gaming Zone

Use Cat6 Ethernet cables with RJ-45 connectors to connect gaming consoles, PCs, and smart TVs to a Gigabit switch or router. Wired Ethernet gives a stable, low-latency connection compared with relying on Wi-Fi alone. Use fiber-optic cable between network switches or to a central server if the distance is long or the venue needs a high-capacity backbone. Use HDMI cables from each console/media player to its screen for video and audio; HDMI is not a network cable.

Recommended connection: Console / media PC — Cat6 + RJ-45 — Switch/Router — Cat6 or fiber uplink — Server/Internet. Console/media PC — HDMI — Screen.

5. Example Photo/Description of a Network Cable in Use

Illustrative example: An Ethernet cable with an RJ-45 plug connects a laptop or Wi-Fi router to a network switch. The clear plastic plug contains eight metal contacts that connect the eight wires to the device's Ethernet port. This is an example illustration, not a photo taken from your home or college. If your teacher requires your own photo, take a picture of the cable behind your router and add it here.

Illustrative Ethernet Cable Connection

Conclusion

Choosing the correct cable and connector improves network speed, reliability, and troubleshooting. Cat6 with RJ-45 is a good choice for short wired connections, while fiber is useful for long, high-speed links.